A Multi-Modal Student Attendance System with Face Recognition and Time-Bound QR Authentication

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***Abstract*— Attendance is an important component of academic management. However, manual attendance may cause loss of valuable class time, errors, and proxy attendance. Face recognition and QR-based methods can mitigate these issues; however, face recognition can be affected by presentation attacks, and QR codes may be shared between students. This work proposes a multi-modal smart classroom attendance and academic management system that uses both face recognition and time-bounded QR authentication. The face module performs face detection, quality filtering, facial embeddings, and presentation-attack detection, whereas the QR module utilizes a rotating HMAC-SHA256-protected challenge, institutional Google authentication, and session/section validation. The proposed platform consists of teacher and student portals, timetable and session management, manual attendance, attendance integrity scoring, analytics, digital-twin based analysis, and assessment/marks management. The developed system was implemented as a web-based platform. It was evaluated using valid and invalid attendance scenarios, and face-processing performance benchmarking. The obtained results highlight the feasibility of employing multi-modal approaches for attendance management supplemented by academic management features in a single platform.**

Keywords— Smart classroom, student attendance, face recognition, QR authentication, anti-spoofing, HMAC-SHA256, attendance analytics, academic management

I. INTRODUCTION

Attendance is a critical element of academic management, aiding on both the monitoring of student attendance and the generation of reliable academic records. Many classrooms track attendance manually by roll calls or written records. While this maintains control by the instructor, it is often time-consuming during teaching activities and can lead to errors, omissions and proxy attendances. This has led to the adoption of alternative automatic approaches including QR codes, RFID and face recognition. Previous works have demonstrated the benefits of such approaches, but each have limitations.

Face recognition is advantageous to students in that they do not have to actively engage with the system in taking attendance. However, this technology can be limited by lighting conditions, face position, image quality and the distance to the camera. Another concern is that of presentation attacks; a photograph or other representation of a student can be used to impersonate their characteristics. QR-based attendance is convenient in that it only necessitates a smartphone device and can be used as a substitute when face recognition is not viable. However, QR codes shown in a classroom can be photographed and shared, requiring additional steps for verification.

This work presents a multi-modal smart classroom attendance and academic management system which combines face recognition and time-bound QR- authentication. The face-based module consists of face-detection, quality filtering, facial embeddings, and presentation-attack detection prior to recording attendance. The QR module consists of a rotating time-bound challenge protected by HMAC-SHA256 plus institutional Google authentication and session/section validation for a variety of authentication options appropriate for classroom settings.

The system is also designed to support for other related activities around attendance management. It offers separate teacher and student portals, classroom and session management, timetable and subject information, manual attendance, attendance reports, integrity scoring, attendance analytics, digital-twin-based attendance analysis, and assessment and marks management. This way, attendance data can be maintained and analyzed as an enabler of wider academic management rather than a standalone record.

It is the aim of this work to unify attendance, authentication, classroom management, and academic management in a single platform. Novelty claims are not made for the technologies presented individually. Instead, this work focuses on their practical application in the classroom and the security considerations involved.

II. LITERATURE SURVEY

A. Face-Recognition-Based Attendance

Face recognition has been widely studied for classroom attendance. Bhattacharya et al. introduced the Smart Attendance Monitoring System (SAMS) that utilizes face detection, face quality, and recognition to automate attendance in a classroom setting [1]. They note the importance of selecting appropriate facial images before performing recognition. Chintalapati and Raghunadh designed an automated attendance system that consists of image acquisition, face detection, preprocessing, feature extraction, classification, and post-processing steps. Their study discusses spoofing as a security concern for face-based attendance systems [2]. Dev and Patnaik studied a face-recognition attendance system that used various recognition approaches and evaluated performance under varying illumination, head movement, facial expressions, and camera distance [3]. These studies show the potential for contactless attendance while also illustrating how recognition performance can be affected by imaging conditions.

Further studies have examined the use of real-time operation in classes. Sultan et al. introduced a real-time university classroom attendance system, which used an HD camera, HOG-based facial feature detection, and the Dlib face-recognition API. Their system was able to recognize multiple students from different viewpoints and achieved an accuracy of 97.38% under their experimental conditions [4]. Lateef and Kamil designed a smart-classroom attendance system with YOLOv7 and a database of 31 students, comparing approaches of machine-learning and deep-learning-based face recognition [5]. Raj et al. also introduced a face-recognition-based smart attendance system to reduce the need for manual attendance [6]. These studies illustrate the shift from traditional face-recognition pipelines to those that allow for real-time and automated attendance in the classroom.

The use of multimodal verification has also been investigated. Akbar et al. combined face recognition and RFID verification for classroom entry and attendance, using facial identities along with an RFID-based identification card [7]. Their system demonstrates the use of multiple independent identification mechanisms within a single attendance procedure. However, such systems entail additional hardware requirements and implementation complexity.

B. QR-Code-Based Attendance

QR codes have been considered as an alternative solution since students can use mobile devices to check their attendance. Liew and Tan proposed a QR-code-based attendance management system that verified the class time, registered mobile devices, and geolocation in addition to QR code [8]. Their findings showed that contextual information could be used to enhance such solutions in order to prevent attendance submissions from unauthorized locations of users. Wiriasto et al. introduced a QR code-based Android attendance application using Flutter that is operated by different features intended for students and lecturers, including geolocation QR-code scanning [9]. Bacuna and Dadiz introduced an eSAM server and mobile QR attendance system that recorded the course name, section, subject, date, and attendance information while providing summary reports to instructors and administrators [10].

In turn, other researchers investigated QR-based attendance management systems that included other tracking and management procedures. Casunuran et al. presented a QR attendance system that included GPS-based student tracking and SMS notification for the arrival and departure of students [11]. Masalha and Hirzallah introduced a QR-based attendance management system for student attendance recording using QR identification [12]. Nuhi et al. developed a QR-based web attendance management system aimed at helping attendance recording and providing automatically generated information on attendance [13]. Pati et al. introduced a QR-based tracking system where personalized QR codes are used by students while their entry and exit information are recorded [14]. Ambaye et al. more recently developed and evaluated an automated QR attendance system for classroom management, reporting improved record keeping, decreased paperwork, and user acceptance [15]..

C. Research Gap

The reviewed literature shows that face recognition can be used to automate student identification, while QR-based systems can ease mobile attendance recording. Some works also included other mechanisms, e.g., RFID checks [7], registered devices and geolocation [8], or location tracking and notifications [11]. Yet, the systems reviewed do not show the same combination of face recognition, presentation-attack detection, time-bounded cryptographically protected QRs, institutional identity checks, and unified classroom and academic management that this work introduces.

As such, the contribution of the proposed system is not a new face-recognition algorithm, QR technology, or cryptography: the originality lies in unifying different attendance mechanisms and the associated security checks in a single classroom platform where face recognition and QR authentication can be used as alternatives and share the same attendance-management infrastructure.
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III. PROPOSED SYSTEM

The proposed system is a smart classroom attendance system aimed at minimizing the efforts related to manual attendance while also maintaining the integrity of the attendance process. Several attendance modes are allowed including QR attendance, face recognition attendance, QR and face recognition attendance, and manual attendance, all housed in one software. Attendance is tied to classroom sessions and timetable, thus enabling reports generation, integrity checks and academic analytics.

A. System Architecture

The system is designed using a client-server model with React.js as the frontend framework plus Django REST

framework as the backend framework. The system has two interfaces, the teachers' and students' interfaces. The backend comprises authentication, classroom sessions, attendance, face data, timetable, and student records.

Initially, the teacher logs in to the system and creates a session for the class he/she wants to take. The teacher selects his/her preferred method of taking attendance, thus initiating the verification process. This involves verifying the students' faces depending on the mode selected by the teacher. The system's backend then validates and authenticates the information provided and registers the attendance of the students in the created session. The records are thereafter subjected to integrity tests, reports generated, and analytics done as explained in Fig.1 .

Authentication → Session creation → Mode selection → Student verification → Validation and authentication → Attendance recording → Integrity analysis → Analytics.

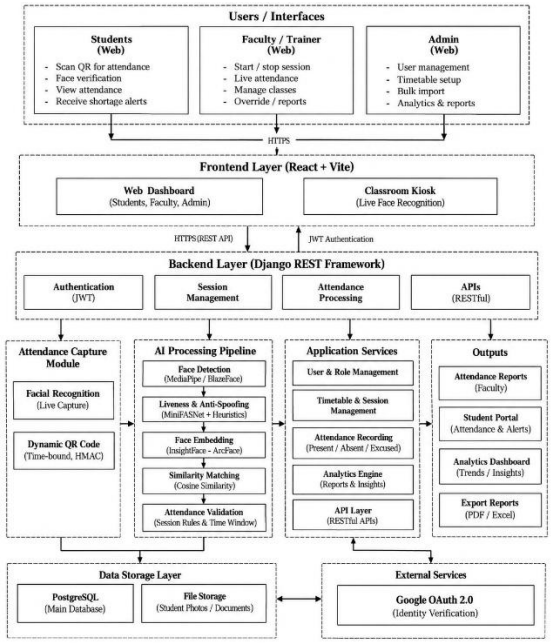


Fig. 1: System Architecture

Module	Main Function
QR Attendance	Time-bound QR challenge and institutional authentication
Face Attendance	Face detection, embedding generation, and identity matching
Presentation-Attack Detection	MiniFASNet ensemble and image-based rejection checks
Attendance Integrity	Verification metadata and 0–100 integrity scoring
Session Management	Class session creation, activation, and attendance control
Academic Management	Subjects, assessments, marks, and student records
Analytics	Attendance summaries, shortages, trends, and Digital Twin analysis

Table I — System Modules and Functions

B. QR-Based Attendance

QR module provides a simple QR code-based attendance recording mechanism with a smartphone without displaying QR code permanently. When a QR session is initiated, the backend generates a unique session token and secret key to create a QR challenge with a HMAC-SHA256 encrypted value with the timestamp window.

Any captured QR codes are invalidated by a new challenge generated every 10 seconds. Students authenticate by logging in with their institutional Google account and scanning the QR code. The backend validates the identity token, matches the student with the corresponding QR session, and records the attendance after successful challenges and Google authentication, as depicted in Fig. 2.

Besides the QR scanning step, the proposed solution offers multiple levels of protection to prevent unauthorized marking of attendance. Simultaneously, it ensures that a student is not marked present more than once for the same session.

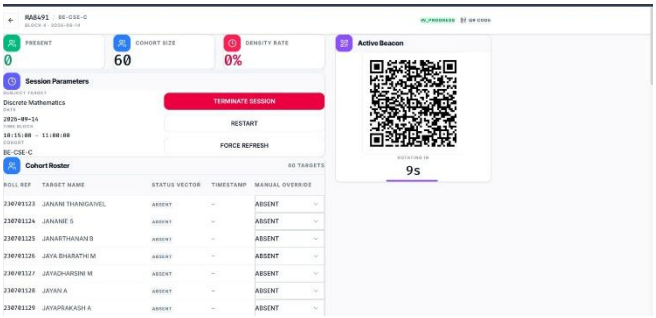


Fig. 2: Dynamic Qr Attendence

C. Face Recognition and Presentation-Attack Detection

This face module is meant to replace QR check-in without touching anything. A camera in the room looks at the student. Then MediaPipe runs the first face search. It keeps only frames that meet simple rules like detection confidence, face size, and where the face sits in the image.

After that, the kept frames go through the InsightFace buffalo_l** setup. From each usable frame, the system makes a feature vector with 512 values. Those vectors are scaled with an L2 step. Next, the system compares each vector to the stored vectors for students in the same section. It uses cosine similarity for that match. The system logs attendance only if the match passes the identity test.

Face matching alone cannot prove that a real person is in front of the camera. A spoofing check is added for that reason. The method uses MiniFASNetV2 and MiniFASNetV1SE together as an ensemble. They estimate if the face looks live. After that, the system runs four quick image tests to reject likely attacks. These are Laplacian texture checking, Canny edge density checking, FFT based moiré checks, and Hough based device border checks. Attendance is accepted only after the system sees a run of valid frames. The full face work flow is shown in Fig. 3.

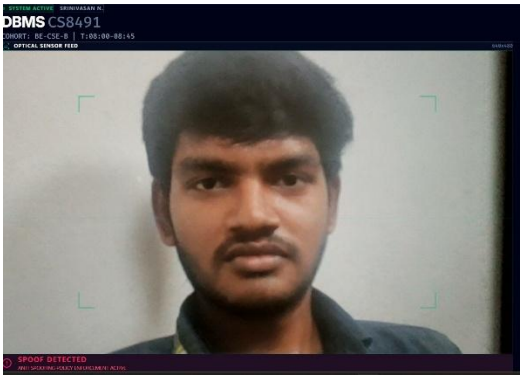


Fig. 3: Anti-Spoofing in Face Recognition.

D. Attendance Integrity

The system stores for each check an attendance state plus additional information. Each student's record contains within a session their ID, time of check, method of verification, face confidence, and additional verification notes.

The attendance state can be PRESENT, ABSENT, LATE, or EXCUSED.

A percentage between 0 and 100 is calculated for each check based on such factors as QR validity, face recognition confidence, anti-spoofing confidence, a check's temporal validity, and the verification path taken. Notes about the score and general verification notes are attached to the check, allowing the teacher to assess the reliability of each attendance record and draw attention to potentially suspicious records.

E. Classroom and Academic Management

The attendance system is integrated with timetable and session management to enable recording of attendance against the relevant class. Through the teacher portal sessions can be managed, attendance rosters and reports viewed and reviewed, integrity information viewed, assessments managed and authorized attendance corrections made.

The student portal provides access to attendance history, timetable, subjects, grades, and personal analytics. This ensures that attendance information is accessible to the students outside the classroom providing a better overview of students' academic standing.

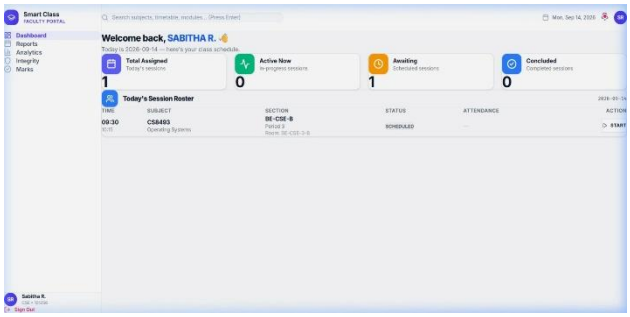


Fig. 4: Faculty/Teacher Page

F. Attendance Analytics and Digital Twin

The attendance data can be further processed to produce student- and class-based attendance summaries, find attendance shortages, and analyze attendance trends within sessions and subjects. Additionally, the system utilizes a Digital Twin to analyze trends in attendance history and determine whether the trends are improving, deteriorating, or stable for a given student.

The module is then able to estimate how many more sessions a student would need to attend to meet the required level of attendance and, subsequently, perform a what-if analysis depending on the projected future attendance levels. Similarly, attendance data can be combined with assessment results to evaluate risk levels.



Fig. 5: Student Portal and Analytics

Conclusively, as seen in the proposal, the system provides attendance verification, classroom session control, attendance audit processes, academic record storage, and analytical tools in a single system. It must be stressed that the main innovation of this project is not in the creation of new algorithms or cryptographic protocols but in the unification of the aforementioned technologies into a singular smart classroom attendance system.

IV.METHODOLOGY & IMPLEMENTATION

The system was implemented as a Web application using the React frontend and Django REST Framework backend. QR authentication, face recognition, attendance verification and the academic services were unified around the session-based attendance management model.

A. Technology stack

The frontend was built using React 18 and Vite while the backend was using Django 5.x and Django REST Framework. The REST APIs unified the system's frontend and backend parts. SimpleJWT was used to provide the application's authentication while the institutional Google authentication was used during the QR-based attendance sessions. SQLite was used for the development while the production used the database. The face processing part used MediaPipe to detect the faces on the client-side, while InsightFace buffalo_l for the face embeddings, while NumPy and SciPy were used for the analytics module.

B. QR authentication

When the teacher begins the QR session, the backend generates a unique session token and a random session secret which are used to build a temporary challenge with the session identifier, time-window and HMAC-SHA256 digest.

QR-based attendance uses a dynamic 10 seconds' time-window which renews itself periodically. When a student scans a QR code, the server validates the challenge, the time-window and decodes the Google institutional ID token to check against the known user profile and attendance session. This way, the session, student and QR code are bound together during the scan to provide additional security. Several verification steps are executed instead of relying on the single QR code scan.

C. Face recognition

MediaPipe BlazeFace is used to initialize the face detection from the camera feed, which is subsequently filtered using the confidence threshold, face size and position on the screen before being subject to recognition. This way, the irrelevant video frames can be discarded earlier the pipeline to reduce the processing overhead.

The detected face is processed using the InsightFace buffalo_l model to extract a 512D L2-normalized embedding, which is compared to the known student embeddings using the cosine similarity. A configured cosine threshold is used to determine the match during the recognition. Additionally, MiniFASNetV2 and MiniFASNetV1SE are used to estimate the liveness of the face with the live probability averaged between the two models. Four image-based checks are also used during the liveness detection to discard non-living faces at an early stage: the texture analysis using the Laplacian filter, Canny edge density analysis, FFT-based moiré analysis and Hough-based device border detection. Finally, three consecutive verification frames need to be successfully processed before the result is accepted.

D. Attendance integrity and system integration

The attendance record binds the student, session, status, timestamp, method, face confidence, and verification metadata together into a single record with an additional integrity score between 0 and 100. The score is computed from the QR validation, face recognition confidence, anti-spoofing confidence, temporal consistency of the timestamp, and the method of verification. The backend Application Programming Interfaces (APIs) are organized around the authentication, sessions, attendance, QR verification, recognition, academic records and analytics. The attendance management follows the teacher and student roles, with the teacher managing the QR sessions, and the student having the view-only access. Additionally, attendance binds the timetable information and the session data together, which allows the system to unify the teacher reports, student history, attendance analytics and Digital Twin analysis.

Overall, the implementation focused on unifying the verification methods around the attendance management and providing additional security during the both QR and face recognition steps, while maintaining the connection to the academic records.

V. RESULTS AND LIMITATIONS

The prototype was evaluated in order to verify the correctness of the described attendance procedures, rejection of invalid requests, and behavior of the face-recognition pipeline. However, limited resources were devoted to biometric evaluation and thus most results are based on functional and security testing rather than statistical analysis of recognition or presentation-attack detection rates.

A. Functional Evaluation

The QR attendance procedure was tested on valid and invalid authentication attempts. A valid request was a valid institutional account with a valid QR challenge. Non-institutional accounts, accounts from another section, invalid QR challenges, inactive sessions, and repeated attendance were rejected. These tests demonstrate that the attendance procedure correctly implements identity, session, section, and attendance-duplication checks on the backend.

B. QR Authentication Evaluation

The QR attendance procedure was tested on altered QR codes in order to evaluate the security of the QR parsing and HMAC verification procedure. Altered QR fields such as the session, time window, and HMAC signature resulted on failed attempts. QR challenge generation is based on a session HMAC secret and time window on the server side while institutional Google accounts serve as an additional authentication factor.

QR code rotation is designed to occur every 10 seconds which limits the usefulness of a compromised QR code. These tests demonstrate that the QR attendance procedure resists casual tampering attempts and the presence of an institution account or a valid QR code is not sufficient for successful authentication.

C. Face Recognition and Presentation-Attack Evaluation

The face recognition procedure was tested as a combined procedure of simultaneous face detection, frame filtering, face embedding generation, face matching, and presentation-

attack detection. The prototype uses MediaPipe for face detection and InsightFace for face embedding generation and comparison to the enrolled student embeddings. Four image-based presentation-attack detection checks are implemented using MiniFASNetV2 and MiniFASNetV1SE with additional temporal checks to filter out frames with consecutive valid face detection.

A single valid face frame is not sufficient for attendance due to additional temporal checks. However, the prototype does not include a benchmark presentation-attack detection dataset for evaluating the false acceptance and false rejection rates. Similarly, the recognition accuracy, false acceptance, and false rejection rates cannot be reported for the prototype due to the lack of a controlled benchmark dataset. Thus, the current evaluation should be considered a functional test of the described face-recognition pipeline.

D. Face-Processing Performance

The face-processing performance was previously reported in the IEEE submission as 180-250 ms/frame for GPU execution and 5000-6000 ms/frame for CPU execution. However, the technical audit of the current prototype code did not identify any benchmark dataset or measurements that could be used to reproduce these results. The code includes time measurements of individual components but no timing statistics or large-scale performance benchmarks.

Thus, the performance of the face-recognition procedure in the IEEE submission cannot be reproduced and these performance indicators should be omitted from the public version. Instead, it is possible to report that face recognition using the GPU is significantly faster than the CPU for all components which can be seen in the comparison of the measured values.

E. Discussion

The evaluation demonstrated that the described system architecture allows for the implementation of multiple attendance procedures in a unified classroom platform. QR attendance procedure provides active institutional account authentication with additional session and time-window checks while face attendance procedure serves as a convenient contactless alternative with presentation-attack detection.

Instead of providing strong statistical evidence of recognition or presentation-attack detection performance, the evaluation presented the functional validity of the described approaches to multi-modal attendance. This conclusion is based on the fact that the prototype has not been benchmarked against a large-scale biometric dataset and cannot provide reliable statistical evidence of recognition performance at this time.

The main contribution of this project is the combination of these two attendance procedures into a single classroom platform rather than the performance characteristics of either one.

F. Limitations

The current prototype does not include a large-scale biometric dataset for evaluating the recognition accuracy, false acceptance rates, false rejection rates, and presentation-attack detection rates. Additionally, the current face-enrollment procedure does not include enough reference samples for each student for accurate recognition. Thus, the validity of the prototype depends on the student's appearance during the face detection procedure.

Changes in lighting, occlusion of the face, change of facial expression, and distance between the camera and the student can reduce the accuracy of the prototype. Similarly, the cosine-similarity threshold that determines whether a face matches a reference has not been calibrated using a large-scale benchmark dataset. The presentation-attack detection procedure similarly has not been benchmarked against a standard benchmark dataset.

The QR attendance procedure has not been stress-tested across multiple classrooms while the current SQLite-based prototype has not been optimized for production traffic. The code-base contains legacy and new attendance procedures while the complete end-to-end QR_AND_FACE procedure has not been tested.

Test Case	**Expected Result**	**Observed Result**
Valid institutional account with valid QR	Attendance accepted	Passed
Non-institutional Google account	Attendance rejected	Passed
Student from incorrect section	Attendance rejected	Passed
Valid QR from active session	Attendance accepted	Passed
Modified or invalid QR challenge	Attendance rejected	Passed
Invalid or closed session	Attendance rejected	Passed
Repeated attendance submission	Duplicate prevented	Passed

Table II. Functional Evaluation Of Qr Attendance

VI. FUTURE WORK AND CONCLUSION

A. Future Work

The future work includes enhancing the system's reliability, evaluation, and scalability. A large and diverse face dataset can be used to evaluate the recognition and presentation-attack mechanisms with different lighting, poses, occlusions, and classroom conditions. Additionally, the threshold of the face matching can be calibrated between genuine and impostor samples.

The QR_AND_FACE workflow needs to undergo more extensive evaluations in multiple classrooms and users to determine its practical performance and scalability. The same applies to the QR_AND_FACE workflow in terms of combined verification mechanism. Apart from that, other possible improvements include enhanced attendance audit logging, optimized face processing, and database migration from the current development database to production-grade database.

B. Conclusion

This paper proposed a smart classroom attendance system that integrates QR-based authentication, face recognition with presentation-attack detection, attendance integrity assessment, and academic analytics into one platform. Unlike the existing solutions that focus only on the mechanisms to record attendance, this study demonstrates how attendance records are connected to classroom sessions, timetable, academic records, and student analytics.

The functional tests revealed that QR workflow is able to accept a valid attendance request and reject a few invalid or unauthorized requests. Similarly, the face-based component can be used as a contactless alternative that is empowered by the presentation-attack detection mechanism. The proposed system is also expanded through teacher and student portals, attendance reports, shortage analytics, and Digital Twin-based insights.

The current proof of concept demonstrated that it is possible to combine all these components in one classroom platform. However, the absence of a large labelled biometric dataset and extensive evaluations in a controlled environment prevent the performance analysis on a larger scale. The future work will include additional validation in different classrooms and users to ensure the system's practical reliability.

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